Beginning AngularJS

Andrew Grant

Apress®

Beginning AngularJS

Copyright © 2014 by Andrew Grant

This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed. Exempted from this legal reservation are brief excerpts in connection with reviews or scholarly analysis or material supplied specifically for the purpose of being entered and executed on a computer system, for exclusive use by the purchaser of the work. Duplication of this publication or parts thereof is permitted only under the provisions of the Copyright Law of the Publisher's location, in its current version, and permission for use must always be obtained from Springer. Permissions for use may be obtained through RightsLink at the Copyright Clearance Center. Violations are liable to prosecution under the respective Copyright Law.

ISBN-13 (pbk): 978-1-4842-0161-9

ISBN-13 (electronic): 978-1-4842-0160-2

Trademarked names, logos, and images may appear in this book. Rather than use a trademark symbol with every occurrence of a trademarked name, logo, or image we use the names, logos, and images only in an editorial fashion and to the benefit of the trademark owner, with no intention of infringement of the trademark.

The use in this publication of trade names, trademarks, service marks, and similar terms, even if they are not identified as such, is not to be taken as an expression of opinion as to whether or not they are subject to proprietary rights.

While the advice and information in this book are believed to be true and accurate at the date of publication, neither the authors nor the editors nor the publisher can accept any legal responsibility for any errors or omissions that may be made. The publisher makes no warranty, express or implied, with respect to the material contained herein.

Managing Director: Welmoed Spahr
Lead Editor: Louise Corrigan
Developmental Editor: Gary Schwartz
Technical Reviewer: Andrew Markham-Davies
Editorial Board: Steve Anglin, Mark Beckner, Ewan Buckingham, Gary Cornell, Louise Corrigan, Jim DeWolf, Jonathan Gennick, Robert Hutchinson, Michelle Lowman, James Markham, Matthew Moodie, Jeff Olson, Jeffrey Pepper, Douglas Pundick, Ben Renow-Clarke, Dominic Shakeshaft, Gwenan Spearing, Matt Wade, Steve Weiss
Coordinating Editor: Christine Ricketts
Copy Editor: Michael Laraque
Compositor: SPi Global
Indexer: SPi Global
Artist: SPi Global
Cover Designer: Anna Ishchenko

Distributed to the book trade worldwide by Springer Science+Business Media New York, 233 Spring Street, 6th Floor, New York, NY 10013. Phone 1-800-SPRINGER, fax (201) 348-4505, e-mail orders-ny@springer-sbm.com, or visit www.springeronline.com. Apress Media, LLC is a California LLC and the sole member (owner) is Springer Science + Business Media Finance Inc (SSBM Finance Inc). SSBM Finance Inc is a Delaware corporation.

For information on translations, please e-mail rights@apress.com, or visit www.apress.com.

Apress and friends of ED books may be purchased in bulk for academic, corporate, or promotional use. eBook versions and licenses are also available for most titles. For more information, reference our Special Bulk Sales–eBook Licensing web page at www.apress.com/bulk-sales.

Any source code or other supplementary material referenced by the author in this text is available to readers at www.apress.com. For detailed information about how to locate your book's source code, go to www.apress.com/source-code/.

I would like to dedicate this book to my wonderful wife, Monica, and my amazing children, Jennifer, Catie and Christopher.

Contents at a Glance

About the Author xiii

About the Technical Reviewer xv

Acknowledgments xvii

■ Chapter 1: JavaScript You Need to Know 1

■ Chapter 2: The Basics of AngularJS 35

■ Chapter 3: Introduction to MVC 47

■ Chapter 4: Filters and Modules 57

■ Chapter 5: Directives 75

■ Chapter 6: Working with Forms 91

■ Chapter 7: Services and Server Communication 115

■ Chapter 8: Organizing Views 131

■ Chapter 9: AngularJS Animation 149

■ Chapter 10: Deployment Considerations 163

Index 177

Contents

About the Author xiii

About the Technical Reviewer xv

Acknowledgments xvii

Chapter 1: JavaScript You Need to Know 1

JavaScript Primer 1

Including Scripts on a Page 1

Statements 2

Functions 4

Parameters and Return Values 5

Types and Variables 6

Primitive Types 8

JavaScript Operators 11

Equality vs. Identity 13

Pre- vs. Post-Increment 15

Working with Objects 17

Creating Objects 17

Reading and Modifying an Object's Properties 18

Adding Methods to Objects 20

Enumerating Properties 21

Control Flow 22

Working with Arrays 25

Callbacks 29

JSON 32

Summary 33

Chapter 2: The Basics of AngularJS 35

Why We Need Frameworks 35

What Is a Framework? 36

Downloading and Installing AngularJS 36

Browser Support 37

Your First AngularJS Application 38

Declarative vs. Procedural Programming 41

Directives and Expressions 41

Summary 45

Chapter 3: Introduction to MVC 47

Design Patterns 47

Model View Controller 51

Why MVC Matters 52

MVC the AngularJS Way 53

Summary 56

Chapter 4: Filters and Modules 57

Introduction to Filters 57

Built-in Filters 60

The Number Filter 60

The Date Filter 62

The *limitTo* Filter 65

AngularJS Modules 68

What Is a Module? 68

Bootstrapping AngularJS 69

Creating a Custom Filter 71

Summary 74

Chapter 5: Directives 75

The Basics of Directives 76

Using Directives 76

Built-in Directives 81

ngBind 81

ngCloak 81

ngInclude 82

ngShow and ngHide 82

ngRepeat 83

Event-Handling Directives 84

Using the API Documentation 84

Creating a Custom Directive 85

The restrict Option 86

The template Option 87

The link Option 87

Summary 90

Chapter 6: Working with Forms 91

HTML Forms Overview 91

The *form* Element 91

The *input* Element 92

The *textarea* Element 94

The *select* Element 95

The *label* Element 96

Model Binding 96

AngularJS Forms 101

Validating Forms 106

Summary 114

Chapter 7: Services and Server Communication 115

Using Services 116

The *$window* Service 116

The *$location* Service 117

The *$document* Service 118

Creating Services 119

Promises 121

Server Communication 122

Handling Returned Data 129

Summary 130

Chapter 8: Organizing Views 131

Installing the *ngRoute* Module 131

Using URL Routes 133

Defining Routes 133

Route Parameters 140

Eager vs. Conservative Routes 145

Route Configuration Options 145

HTML5 Mode 147

Summary 148

Chapter 9: AngularJS Animation 149

Installing the *ngAnimate* Module 149

CSS Animation Overview 150

Transforms 151

Transitions 153

Applying Animations 156

Summary 161

Chapter 10: Deployment Considerations 163

Configuration 163

Testing 165

Error Handling 167

Hide Unprocessed Templates 169

Minification and Bundling 170

Managing the Build Process 171

Deployment 173

Summary 176

Index 177

About the Author

Andrew Grant has been developing websites for fun and profit for over 18 years. He is currently a full-time Web Developer with a passion for technology and life on the cutting edge. Andrew holds various web development related vendor certifications and an Advanced Diploma in Software Development. He can be reached via his personal web page at http://andrewgrant.net.au.

About the Technical Reviewer

Andrew Markham-Davies is a software engineer from Yorkshire, England who has worked on software for the like of Sky, BBC, ADT and DePuy Synthes. While he's has worked extensively with C#, java and php in the past, his passion is Javascript and front end development. A big fan of mv* style frameworks, Andrew found AngularJS the perfect tool for building rich, well structured javascript heavy applications.

He is a keen advocate of refactoring and applying engineering principals to Javascript including automation, tooling and testing best practices.

Andrew can be found on twitter at http://twitter.com/atmd83 and blogs at http://atmd.co.uk.

Acknowledgments

I would like to thank my wonderful wife, Monica, for helping me write this book and do my day job at the same time. I would also like to thank the amazing Apress editorial team, in particular my technical reviewer, Andrew Markham-Davies, for his valuable feedback.